Question Paper: Machine Learning

Machine Learning Question Paper

Instructions:

* The question paper consists of three sections: Multiple Choice Questions, Short Questions, and Long/Descriptive Questions.
* Attempt all questions in the respective sections.
* Use a blue or black pen to write your answers.
* Do not write your name or roll number on the answer sheet.

Section A: Multiple Choice Questions (30 marks)

1. What is the primary goal of machine learning?

A) To develop algorithms for solving mathematical problems

B) To develop models that can learn from data and make predictions

C) To develop software that can interact with humans

D) To develop hardware that can process large amounts of data

Correct answer: B) To develop models that can learn from data and make predictions

2. Which of the following machine learning algorithms is used for classification problems?

A) Linear Regression

B) Decision Trees

C) Clustering

D) All of the above

Correct answer: B) Decision Trees

3. What is the term for the process of dividing a dataset into training and testing sets?

A) Data preprocessing

B) Data visualization

C) Data splitting

D) Cross-validation

Correct answer: C) Data splitting

4. Which of the following is a type of supervised learning?

A) Unsupervised learning

B) Reinforcement learning

C) Regression

D) Clustering

Correct answer: C) Regression

5. What is the purpose of regularization in machine learning?

A) To increase the complexity of the model

B) To decrease the complexity of the model

C) To improve the accuracy of the model

D) To reduce the training time of the model

Correct answer: B) To decrease the complexity of the model

6. Which of the following algorithms is used for clustering?

A) K-Means

B) K-Nearest Neighbors

C) Linear Regression

D) Decision Trees

Correct answer: A) K-Means

7. What is the term for the measure of how well a model is able to predict the target variable?

A) Accuracy

B) Precision

C) Recall

D) Coefficient of determination

Correct answer: D) Coefficient of determination

8. Which of the following is a type of neural network?

A) Decision Trees

B) Random Forest

C) Convolutional Neural Network

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D) Support Vector Machine

Correct answer: C) Convolutional Neural Network

9. What is the purpose of feature scaling in machine learning?

A) To improve the accuracy of the model

B) To reduce the complexity of the model

C) To prevent overfitting

D) To prevent features with large ranges from dominating the model

Correct answer: D) To prevent features with large ranges from dominating the model

10. Which of the following is a popular library for machine learning in Python?

A) NumPy

B) Pandas

C) Scikit-learn

D) TensorFlow

Correct answer: C) Scikit-learn

Section B: Short Questions (15 marks)

1. What is the difference between supervised and unsupervised learning? (3 marks)

2. Explain the concept of overfitting in machine learning. (3 marks)

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3. What is the purpose of feature engineering in machine learning? (3 marks)

4. Describe the steps involved in building a machine learning model. (4 marks)

5. What is the difference between a neural network and a decision tree? (4 marks)

Section C: Long/Descriptive Questions (30 marks)

1. Explain the concept of deep learning and its applications in machine learning. (10 marks)

2. Describe the steps involved in building a predictive model using linear regression. (10 marks)

3. Discuss the importance of data preprocessing in machine learning and explain the different techniques used. (10 marks)

Answer Key

Section A: Multiple Choice Questions

1. B) To develop models that can learn from data and make predictions

2. B) Decision Trees

3. C) Data splitting

4. C) Regression

5. B) To decrease the complexity of the model

6. A) K-Means

7. D) Coefficient of determination

8. C) Convolutional Neural Network

9. D) To prevent features with large ranges from dominating the model

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10. C) Scikit-learn

Section B: Short Questions

1. Supervised learning involves training a model on labeled data, while unsupervised learning involves training a model on unlabeled data.
2. Overfitting occurs when a model is too complex and performs well on the training data but poorly on new, unseen data.
3. Feature engineering involves selecting and transforming raw data into features that are useful for building a machine learning model.
4. The steps involved in building a machine learning model include data collection, data preprocessing, feature engineering, model selection, training, and evaluation.
5. A neural network is a type of machine learning model that is composed of multiple layers of interconnected nodes, while a decision tree is a type of machine learning model that uses a tree-like structure to make predictions.

Section C: Long/Descriptive Questions

1. Deep learning involves the use of neural networks with multiple layers to learn complex patterns in data. Applications of deep learning include image and speech recognition, natural language processing, and game playing.
2. The steps involved in building a predictive model using linear regression include data collection, data preprocessing, feature engineering, model selection, training, and evaluation. The goal of linear regression is to learn a linear relationship between the input features and the target variable.

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3. Data preprocessing is an important step in machine learning because it can significantly impact the performance of the model. Techniques used in data preprocessing include handling missing values, scaling and normalizing features, and encoding categorical variables.